

SCI-MAP

# TolTEC Ordinary Mapmaking and Observation Coaddition

## Scientific Basis, Response, and Validity

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|                          |                                                                                             |
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This document explains the map estimator, its response and uncertainty, the meaning of support and validity, and the scope of observation coaddition for TolTEC scientists. The separate formal contract remains the normative authority for the 52 requirements, 25 predictions, exact states, provenance, and conformance procedure.

Scientific owner: Grant Wilson

## Executive summary and current status

For a map pixel  $p$ , SCI-MAP forms the coefficient-weighted average

$$\hat{m}_p = \frac{\sum_i G_{pi} \gamma_i z_i}{\sum_i G_{pi} \gamma_i} = \frac{N_p}{Q_p}, \quad (1)$$

using only realized PTC samples that are available under PTC’s declared identity and scope, have valid ALIGN/AST coordinates, and pass MAP’s admission conditions. Here  $G_{pi}$  describes how sample  $i$  is placed in its unique containing pixel, and  $\gamma_i$  is the exact PTC-owner-selected MAP-facing coefficient. The admitted Registry contains the uniform family `SCI-PTC:uniform_constant@draft-0.1`, with exact dimensionless  $\gamma_i = 1$ , but Registry presence and MAP permission do not select or realize it. A numerical route requires explicit selection, complete PTC publication, exact target/payload/occurrence association, a requested, applicable, eligible, realized `SCI-PTC:uniform_coefficient_handoff@draft-0.1` decision, independent MAP admission, and MAP finite-positive classification. There is no default or fallback. Pixels without adequate valid support are unavailable; they are not measurements of zero sky brightness.

For compatible observation maps on one common grid, the v0.2 coadd is

$$\hat{s}_p = \frac{\sum_o u_{op} \hat{m}_{op}}{\sum_o u_{op}}, \quad (2)$$

where the named v0.2 uniform family sets dimensionless  $u_{op} = 1$  for every admitted observation row. This is equal-observation averaging, not precision. An observation bundle is admitted atomically: its signal, coordinates, response, uncertainty, validity, and provenance must be mutually compatible before any of its valid pixels contribute. Pixel-level support then determines where the admitted observation contributes.

Both equations are linear only *conditional on fixed* sample and detector membership, projection, coefficients, support policy, response state, geometry/WCS, calibration, and upstream processing. If any of those states is estimated from the same data, unconditional bias or covariance requires a joint model or resampling result.

|                                           |                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Intended scientific estimator</b>      | Positive-coefficient ordinary map and compatible common-grid observation coadd, Equations (1)–(2).                                                                                                                                                                                                                                                                                      |
| <b>Upstream input status</b>              | MAP always consumes a realized positive-rank PTC r0.5 product; there is no CAL fallback, PTC-disabled MAP route, or inferred no-op PTC product. The admitted uniform family can fill the coefficient slot only through explicit selection and its complete exact handoff. The approved finite-nonnegative <code>coverage_cut</code> policy supplies no universal value or mode default. |
| <b>Implementation conformance</b>         | Not assessed in this authorship revision.                                                                                                                                                                                                                                                                                                                                               |
| <b>Response/representation validation</b> | Not established; response injections and WCS round trips remain required evidence.                                                                                                                                                                                                                                                                                                      |
| <b>Observational validation</b>           | Not established; amplitude, shape, position, covariance, and repeatability require astronomical controls.                                                                                                                                                                                                                                                                               |
| <b>Production readiness</b>               | Not claimed and cannot be inferred from the algebraic contract.                                                                                                                                                                                                                                                                                                                         |

| Route level                                        | Status                                            | Principal remaining gate                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conditional estimator<br>Numerical observation map | Defined<br>CANDIDATE; not<br>source-closed/frozen | Equations and scientific meanings only.<br>Exact selected uniform-family handoff or<br>another future admitted family, valid<br><b>coverage_cut</b> state/value, every independent<br>gate, and owner-accepted/canonically<br>activated r0.2 source/profile binding. No<br>realized application route is claimed.                 |
| Response-bearing map                               | Product-dependent                                 | MAP-local realized operator is always<br>disclosed; the selected role requires exactly the<br>fixed-state or full-procedure response object it<br>names and its complete identity/comparison<br>record. Absence of a different family does not<br>block it. Whole-chain response is unavailable<br>unless separately established. |
| Covariance-qualified map                           | Product-dependent                                 | Exact required covariance representation and<br>every block named by the selected role.                                                                                                                                                                                                                                           |
| Compatible-grid coadd                              | Conditional; route<br>binding CANDIDATE           | Exact admitted observation maps and<br><b>observation_coadd_admission@2</b> ; companion<br>qualification remains role-specific. No<br>source-closed frozen route is claimed.                                                                                                                                                      |
| Reprojection/mosaic                                | Not authorized                                    | Separate owner decision and contract.                                                                                                                                                                                                                                                                                             |

The exact MAP lifecycle is **requested**, **effective**, **observation\_resolved**, **applied**, **realized**, in that order. **requested** is immutable accepted intent. **effective** is its supported executable interpretation. **observation\_resolved** means every observation-specific parent, coefficient, profile, geometry, mapping, parameter, support rule, output role, and persistence decision is concrete or typed unavailable. **applied** binds the exact resolved operation and parameter state to the invocation or commitment of one execution attempt. **realized** binds the immutable resulting accumulators, products, statuses, failures, and provenance that were actually produced. A stop before application has neither an applied nor a realized record. An applied attempt that terminates before output realization retains its applied-attempt identity and exact terminal failure but creates no realized product.

| MAP stage                   | Exact meaning                                                                                                                                                                         | Record rule                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>requested</b>            | Immutable accepted intent.                                                                                                                                                            | Exists only for an accepted request; it is<br>never reconstructed from a later stage.                                                                             |
| <b>effective</b>            | Supported executable interpretation of that<br>request.                                                                                                                               | Binds supported/disabled/unavailable<br>interpretation without mutating intent.                                                                                   |
| <b>observation_resolved</b> | Every observation-specific parent, coefficient,<br>profile, geometry, mapping, parameter, support<br>rule, output role, and persistence decision is<br>concrete or typed unavailable. | Supplies the exact value/state for<br>pre-application validation.                                                                                                 |
| <b>applied</b>              | Exact resolved operation and parameter state<br>was invoked or committed for one execution<br>attempt.                                                                                | Created only at the application event;<br>retained with exact terminal failure<br>when output realization fails.                                                  |
| <b>realized</b>             | Immutable resulting accumulators, products,<br>statuses, failures, and provenance were actually<br>produced.                                                                          | Created only for actual realization; it<br>never authorizes its own earlier<br>application.                                                                       |
| Termination rule            | Stop before application, or stop after<br>application but before output realization.                                                                                                  | The former has neither applied nor<br>realized record; the latter has an<br>applied-attempt record and terminal<br>failure but no fabricated realized<br>product. |

The MAP plan, coefficient selection/application, projection, support policy, output/coadd plan, response/covariance request, product role, and **coverage\_cut** use this same ordering. Each parameter stage binds its exact value and provenance; a completed realization contains all five stages, while an earlier stop contains only stages actually reached.

The MAP lifecycle does not replace the independently named PTC/VAL handoff axes **requested/not\_requested**, **applicable/inapplicable/applicability\_unknown**, **eligible/ineligible/decision\_unavailable**, and

**realized/incomplete/failed/not\_produced**. It also does not replace the ECS requirement-result vocabulary or evidence- artifact realization state. In particular, a MAP parameter’s resolved or applied value may be checked before application or bound after application; its **realized** value is never a prerequisite for a pre-application check.

Resolution, application, and realization are distinct. Pre-application validation examines the exact **observation\_resolved** parameter and profile state. Application binds that resolved state to one attempt. A later realized record binds what that attempt actually produced; it cannot be used retroactively as the prerequisite that authorized the attempt.

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# 1 What SCI-MAP estimates

SCI-MAP estimates a map-domain average of the *realized PTC-transformed* calibrated- $x$ -derived nonpolarimetric total-intensity-equivalent quantity  $z_i = Z_i^{\text{PTC}}$ . Neither `mJy/beam` nor a STOKES token promotes it to Stokes I. It does not estimate unconvolved sky truth. The physical meaning of the result is therefore inherited from the complete upstream CAL-to-PTC chain: what the ordinary `xs` sample represents, how it was calibrated and transformed, what sky coordinate it has, and whether the required PTC product exists under its declared producer-local validity.

The numerical signal scale is the declared top-of-atmosphere, point-source-equivalent `mJy beam-1` convention inherited from SCI-CAL with the exact originating fixed-nominal-beam/template identity. MAP does not create absolute calibration, choose a beam convention, or complete an unresolved broadband photometric definition. A numerical map may consequently exist while literal point-source-peak interpretation remains unavailable because the response for the declared processing chain is incomplete. The same caution applies to extended-source fidelity, integrated flux, and absolute offsets.

For one exact frozen upstream and MAP state, the linear response is the fixed-state sky-to-PTC operator followed once by the map operator. Writing the selected processed-sample vector as  $\mathbf{z} = (z_i)$ ,

$$A_{\text{MAP},\Pi} \equiv A_{\text{out}} = D_{Q,\text{out}}^{-1} J_{\text{out}} G \Omega, \quad \widehat{\mathbf{m}} = A_{\text{MAP},\Pi} \mathbf{z}, \quad R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta}, \quad (3)$$

where the one MAP operator binds the admitted PTC-sample input domain, support-authorized output rows, coefficient generation, one-hot projection, support policy, WCS/grid, parents, and lifecycle generation, while  $H_{\text{fixed},\Theta}$  binds source domain, processed-sample codomain, units, parent, membership, support, coefficient state, and response basis. Preserving a constant *sample vector* does not prove preservation of a constant sky mode: an upstream baseline, common-mode removal, or filter may already have removed it.

A stored kernel is a fixed-state realized response only when its template, sample-domain parent, and processing lineage prove Equation (3). Otherwise it is a tracer or has an explicit limited or unavailable status. That state does not invalidate the numerical map or prohibit later analysis; it only limits the claims made by the original bundle. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity.

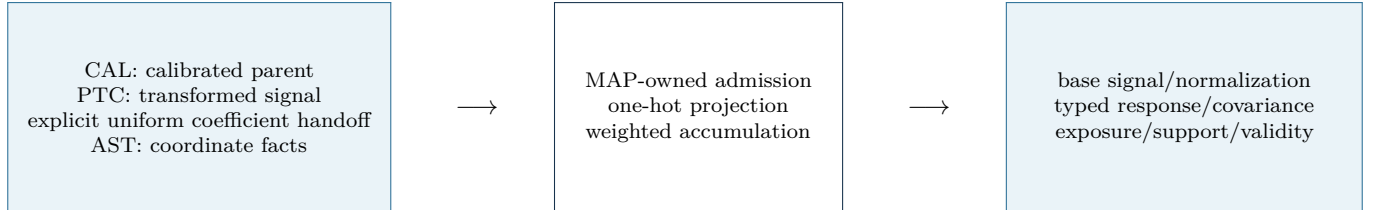


Figure 1: SCI-MAP transforms declared upstream quantities; it does not repair or reinterpret them.

## 2 From samples to pixels

An occurrence  $i$  is not merely time and detector. It binds the observation, detector occurrence and UID, stable RTC output sample  $n$ , exact PTC product/application generation, segment, and array/network/group. Time is an attribute; input-column, row, and container positions are locators. The AST coordinate is exactly `SCI-AST:rtc_output_grid_coordinates@1` for the same  $n$  and parent chain. MAP never joins signal and coordinate by time, row, cardinality, or numerical coordinate equality.

MAP owns `SCI-MAP:one_hot_containing_pixel@1`. A finite coordinate inside the target extent belongs to the unique pixel whose cell contains it under half-open lower-inclusive, upper-exclusive bounds. There  $G_{pi} = 1$ ; it is zero for every other pixel. An internal upper edge belongs to the adjacent cell, while the outer upper boundary and every out-of-grid coordinate contribute nowhere. Fractional projection is deferred until a complete kernel, support, normalization, conservation, boundary, response, and covariance family is separately authorized. AST may materialize  $G$ , but the artifact must bind both the AST coordinate parent and the exact MAP plan.

PTC owns the MAP-facing coefficient  $\gamma_i$ . Frozen PTC r0.5 left its PTC-OD-010 family slot open; the later admitted uniform authority fills one allowed slot with the exact dimensionless value  $\gamma_i = 1$ . The family is usable only after explicit selection, complete atomic PTC publication, exact target/payload/occurrence association, MAP permission, and a requested, applicable, eligible, realized named handoff under the compatible source binding. Registry presence alone selects or realizes nothing, and MAP does not infer unity, loading, sensitivity, scatter, or inverse variance as a fallback. A later coefficient generation does not modify an earlier transformed product, and cross-generation use needs an explicit compatibility decision. Coefficient availability establishes the family and generation, index/broadcast relation, transformed-product compatibility, readable payload presence, typed state, and cause; it does not claim finiteness. MAP alone classifies an authorized payload as finite strictly positive, exact zero, negative finite, non-finite, or unrepresentable.

The representation-independent handoff is `SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4`. MAP owns the occurrence-level policy `SCI-MAP:map_upstream_admission@2`; the exact manifest-bound SCI-VAL Registry revision binds it and VAL Core evaluates it. The policy requires the exact PTC output-retention permission and preserved ancestry, excludes direct synthesized/replaced representatives and unresolved required facts for this use, carries inherited influence without turning it into a universal veto, and records response and uncertainty roles. Pixel placement and final contribution remain MAP estimator actions.

For an exact compatible selected coefficient family the effective contribution is  $G_{pi}\gamma_i$ . For the admitted uniform family this becomes  $G_{pi}$  only after complete PTC publication, exact association, the four-axis handoff, separate MAP admission, and MAP finite-positive classification; the literal value one is not a fallback. Membership is the conjunction of ten separately typed Boolean pass predicates derived from ten retained typed gate records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC proposition; MAP upstream admission; AST coordinate validity; finite transformed signal; finite positive coefficient; admitted projection/boundary; and MAP-local contribution plus required companions. Unknown, conflict, ineligible, unavailable, invalid, and not-requested all project to estimator nonmembership where applicable while their full T/F/U/C or request/applicability/eligibility/realization record, immutable reason, causes, provenance, and failure scope remain intact. For every gate with named request, applicability, eligibility, and realization fields, only requested/applicable/eligible/realized passes. The PTC coefficient/QC pass says only that this field combination holds under the exact PTC-owned coefficient/QC profile; it cannot grant or rescue MAP admission. Evaluation is ordered: structural binding; safe retrieval and classification of only structurally admitted  $z_i$  and  $\gamma_i$ ; coordinate, placement, and companion gates; membership; then accumulation. All gates pass before payload arithmetic. The payload is never screened by multiplication with a numeric zero mask. MAP then accumulates

$$N_p = \sum_{i \in C_p} G_{pi}\gamma_i z_i, \quad Q_p = \sum_{i \in C_p} G_{pi}\gamma_i. \quad (4)$$

The numerator is not yet a sky estimate, and the denominator is a gridding normalization by default. Division occurs only where the adopted support policy permits a scientific map value.

## 3 Projection, response, and removed modes

The one-hot family closes the initial  $G_{pi}$  route without making a fractional-kernel claim. It also keeps geometry distinct from eligibility: AST-geometric incidence can exist when the PTC signal is unavailable; MAP-route-candidate incidence additionally requires the realized route and upstream admission; estimator contribution adds every remaining

numerical and MAP-local gate.

Fixed-state, PTC full-procedure, and PTC+MAP re-resolved response are different scientific objects. The fixed-state family uses Equation (3) with every upstream and MAP choice frozen. The two full-procedure families are defined by the shared equations:

$$\begin{aligned} (\mathcal{D}_{\Delta}^z, \iota_0^z, \iota_+^z) &= \text{authorize}_{\mathcal{C}_{\Delta z}}(z_0, z_+), \\ \Delta \mathbf{z}_{\text{PTC-FP}}(\delta \mathbf{s}; \alpha) &= \iota_+^z \mathcal{P}_{\text{PTC}}(\mathbf{s} + \delta \mathbf{s}; \alpha) - \iota_0^z \mathcal{P}_{\text{PTC}}(\mathbf{s}; \alpha), \end{aligned} \quad (5)$$

$$\Delta \mathcal{S}_{\text{PTC-FP}} = \text{state}_{\text{PTC}}(\mathbf{s} + \delta \mathbf{s}; \alpha) \ominus \text{state}_{\text{PTC}}(\mathbf{s}; \alpha), \quad (6)$$

$$\begin{aligned} (\mathcal{D}_{\Delta}^m, \iota_0^m, \iota_+^m) &= \text{authorize}_{\mathcal{C}_{\Delta m}}(m_0, m_+), \\ \Delta \mathbf{m}_{\text{PTC-FP}} &= \iota_+^m A_{\text{MAP}, \Pi} \mathbf{z}_+ - \iota_0^m A_{\text{MAP}, \Pi} \mathbf{z}_0, \quad \Pi \text{ fixed}, \end{aligned} \quad (7)$$

$$\Delta \mathbf{m}_{\text{PTC+MAP-RR}} = \iota_+^m \mathcal{M}(\mathcal{P}_{\text{PTC}}(\mathbf{s} + \delta \mathbf{s}; \alpha); \Pi_+) - \iota_0^m \mathcal{M}(\mathcal{P}_{\text{PTC}}(\mathbf{s}; \alpha); \Pi_0), \quad (8)$$

A numerical full-procedure difference is defined only by a complete immutable comparison-space record. The record binds baseline and perturbed product identities; scientific quantity and unit; occurrence or MAP output-row identities; source and target frame/WCS; additive reference, gauge, and null- space state; baseline and perturbed support; coefficient and response basis; one exact common comparison domain; an exact map from each result into that domain; explicit treatment of rows present in only one result; perturbation amplitude, sign, side, location, source state, and normalization; finite- difference convention; state-transition record; and numerical availability with cause. It authorizes no implicit zero fill, union-domain sentinel, support intersection, coordinate-nearest or same-shape/row-position association, interpolation, reprojection, or gauge alignment. Without the complete authorized record the numerical difference is unavailable, or the result is only a separately typed state transition/discontinuity.  $\Delta \mathcal{S}_{\text{PTC-FP}}$  never establishes subtractability by itself.

Thus two arrays are never subtracted merely because both procedures produced them. The parameter  $\alpha$  binds amplitude, sign, perturbation side, source state, location, normalization, domain, and every family parameter. The separate  $\Delta \mathcal{S}_{\text{PTC-FP}}$  records changes in masks, groups, rank, loading subspace, classifications, coefficient state, stop cause, and support, but does not prove the payloads subtractable. Holding MAP state fixed still requires exact baseline/perturbed row identities and maps into common PTC and MAP comparison domains. Re-resolving MAP can change  $\Pi_0$ ,  $\Pi_+$ , membership, and  $\mathcal{S}_{\text{out}}$ ; exact common grid and row maps are prerequisites to its numerical difference. The name whole-chain is reserved for a separately authorized RTC-to-CAL-to-PTC-to-MAP rerun from exact immutable parents; that response is unavailable and not claimed here.

The realized MAP operator  $A_{\text{MAP}, \Pi}$  itself is always known and disclosed for a valid MAP product. Its exact PTC-sample domain, support-row codomain, membership, one-hot projection, selected coefficient generation, support policy, WCS/grid, parents, and lifecycle are the MAP-local response statement. This disclosure remains meaningful when the upstream-conditioned operator or procedure response is limited or unavailable; it is not a fabricated upstream or whole-chain response.

A realized PTC-grid companion starts at MAP and receives the fixed MAP operator exactly once. If any signal member lacks a required exact companion, MAP keeps the complete signal membership and reports response limited or unavailable; it does not construct a smaller hidden subset.

The PTC nonrestored additive reference  $\lambda$ , fitted correlated removed component  $\hat{U}_{\text{corr}}$ , total removed component  $U_{\text{total}, \Theta}$ , realized removed-subspace identity, fixed-state null space, and any full-procedure invariant/unidentifiable modes are separate facts. They are not automatically residual bias. The fixed-state perturbation statement is Equation (3); v0.2/r0.2 makes no unnamed  $\mu_0$  or residual conditional-bias claim.

## 4 Weights and uncertainty

For fixed upstream and map state, uncertainty propagates through the same map operator as the signal:

$$C_m = ANA^T, \quad (9)$$

where  $N$  is the declared PTC-domain processed-sample noise covariance when it is available for the asserted domain. Equation (9) retains cross-pixel terms created by shared samples and any correlations already present in the timestream.

These quantities answer different questions:

1.  $Q$ : the gridding denominator that normalizes the map;
2. a formal marginal inverse variance: valid only under its stated coefficient and covariance assumptions;
3. the full conditional map covariance or precision, including correlated pixels when available; and
4. NOI's exact `conditional_detector_sign_randomization_marginal_second_moment` and conditional scale;
5. NOI standardized signal, with only its exact conditional-scale claim; and
6. any future calibrated-significance method, which requires separate authority and remains unavailable here.

SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps  $Q$  normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims.

The base/unfiltered MAP bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, recoverable from lineage, summarized, incomplete, or unavailable. Missing correlations and nuisance terms are not zeros. Normalization, hits, exposure, and diagnostic weight are not covariance. Absence of a complete covariance model neither invalidates the map nor prohibits later analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity.

The distinction is especially important when sample membership, weights, projection, response, or support is estimated from the same observation. The linear equations remain useful conditional descriptions, but they do not include selection uncertainty. A joint model, resampling study, or other separately justified treatment is needed for an unconditional statement.

| Quantity                                 | Owner/status                                   | Science interpretation                                                                                                                                                                 |
|------------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analysis coefficient $\gamma$            | PTC; uniform family admitted, handoff explicit | Exact $\gamma = 1$ only after selection, complete publication, exact association, MAP permission, and realized handoff; no fallback and never precision.                               |
| Normalization $Q$                        | MAP                                            | Denominator of the ordinary estimator; not automatically uncertainty.                                                                                                                  |
| Conditional covariance                   | PTC input, MAP propagation                     | Fixed-state error covariance through projection and normalization.                                                                                                                     |
| Conditional marginal second moment/scale | SCI-NOI                                        | Exact NOI empirical- uncertainty product and conditional scale; no physical-noise, total- uncertainty, complete-covariance, precision, weight, or significance claim.                  |
| Standardized signal                      | SCI-NOI                                        | MAP signal standardized by the stated conditional detector-sign-randomization second-moment scale; no significance, probability, false-rate, completeness, purity, or catalogue claim. |
| Future calibrated significance           | Unavailable                                    | Requires a separately approved exact method; no current MAP or NOI product is promoted to it.                                                                                          |
| Calibration/response nuisance            | SCI-CAL and response owner                     | Retained as quantified, lineage-resolvable, or unavailable; never silently zero.                                                                                                       |

Table 2: Ownership prevents a convenient numerical product from acquiring a stronger statistical meaning.

## 5 Coverage, support, and validity

The formal contract keeps the following facts separate. For science users they answer the following questions:

| Science question                                                           | Product                                       | Interpretation                                                                                                   |
|----------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Did a valid AST coordinate reach this pixel?                               | AST-geometric incidence                       | Geometry independent of PTC signal availability.                                                                 |
| Did an occurrence have the realized PTC route and pass upstream admission? | MAP-route-candidate incidence                 | Route/use policy before local numerical gates.                                                                   |
| How many admitted weighted occurrences entered the estimator?              | Estimator contribution                        | Exact contribution count.                                                                                        |
| How many complete observations entered this coadd pixel?                   | Contributing observation count                | Observation bundles, not samples.                                                                                |
| Which unique original acquisition footprints were upstream eligible?       | Upstream-eligible original-footprint exposure | Own-coordinate footprint accounting; not complete temporal support.                                              |
| Which unique original acquisition footprints were retained?                | Retained original-footprint exposure          | Retained footprint accounting; not precision or effective integration time.                                      |
| Was the denominator large enough to normalize?                             | Numerical support                             | Lower support threshold.                                                                                         |
| Is the pixel inside the adopted science region?                            | Science-policy support                        | Stricter policy threshold.                                                                                       |
| Is the base MAP value scientifically valid under MAP’s own contract?       | MAP-local base-product validity               | Required conjunction of identity, support, finite value, and no MAP-product failure; not downstream eligibility. |

Hits are not validity, exposure is not precision, and positive normalization is not sufficient for final validity. A rectangular FITS image may contain storage values outside the scientific domain; the explicit support and validity products determine which pixels are measurements. Unsupported pixels are unavailable, not zero sky.

Exposure begins with ALIGN-owned  $e_{\text{acq}}$  and  $e_{\text{vo}}$ , measured in seconds on stable original occurrences. RTC, CAL, and PTC preserve those facts, lineage, and causes. MAP deduplicates by observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation. Under `SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1`, it places each positive- $e_{\text{vo}}$  original exactly once using that original’s own AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS and the one-hot half-open rule. Geometry/pointing realization, frame, coordinate validity, boundary state, parent, and version are part of the identity. A descendant RTC-output coordinate is never substituted.

This is deliberately different from causal influence. One original may influence several RTC/PTC outputs at different map pixels; the influence graph is retained, but its edges neither relocate nor duplicate physical seconds. Upstream-eligible exposure uses originals ancestral to route candidates; retained exposure restricts the original population to ancestors of admitted MAP outputs. Under one-hot placement, one original occupies at most one pixel, or none after boundary loss. Coadded exposure unions observation-scoped original identities after atomic admission. Replacement, synthesis, donor reuse, overlapping filters, decimation, duration, cadence, sample count, hits,  $Q$ ,  $\gamma$ , and formal precision cannot create or reconstruct exposure. Missing original lineage or own-coordinate authority makes the exposure role unavailable. These products map unique original acquisition footprints; they are not complete temporal support, effective map integration time, precision, or normalized-map influence. Causal influence and representative RTC occurrence are published separately.

**Approved operational two-threshold policy.** A lower threshold permits numerically safe normalization; a stricter threshold defines the adopted science region. Both are scaled from an order statistic of the finite positive normalizations in a declared map population. This is adopted policy, not a law of mapmaking or a proof of exposure, precision, significance, completeness, response, noise quality, or physical optimality.

Because the reference normalization is computed from a declared population, changing the map extent or population

can change the science mask even when the local samples near a particular pixel are unchanged. That global behavior must be justified and validated for the intended use.

The formal equations compare  $Q_p$  with  $Q_\star c/10$  and  $Q_\star c$ . The dimensionless scalar  $c = \text{coverage\_cut}$  follows requested, effective, observation-resolved, applied, and realized stages. Pre-application validation checks the exact resolved value; application binds it to one attempt, and realization records only what that attempt produced. It is finite and nonnegative. Exact zero requests no relative cut: both relative thresholds are zero, while each row still requires its own finite strictly-positive  $Q_p$ . Values  $0 < c \leq 1$  are the ordinary recommended range. A value above one is admissible only when the independently authorized, same-value and same-scope Boolean plan fact `coverage_cut_above_one_override_requested` is true; the number cannot authorize itself and may intentionally leave science support empty. That record binds request identity, authorized owner/profile, scope, each reached lifecycle stage, exact value, cause, and provenance. A negative, non-finite, missing, or unrepresentable value fails before support rows are constructed, application begins, or a required product is mutated. A completed realization retains all five stages; an earlier stop has no future-stage placeholder. Neither this contract nor the admitted uniform coefficient family supplies a universal numerical or mode default.

Both comparisons are inclusive and retain the inherited exact population and order statistic. Thus a row exactly at either threshold is included in that support plane. An empty policy population gives  $Q_\star = 0$  but creates no supported row without finite positive normalization.

## 6 Observation coaddition

V0.2 combines observation maps only when they represent the same declared quantity on one compatible pixel grid. Each observation first exists as a complete immutable logical bundle: signal, normalization, MAP-local operator, upstream-conditioned response state, covariance status, exposure, support, validity, WCS, lifecycle, identity, and provenance are checked together. Physical persistence is controlled by the effective output plan. A conforming in-memory bundle and its faithfully persisted form have identical admission, scientific identity, and arithmetic. A later coadd-only entry may read persisted conforming bundles without rerunning upstream processing. Every plan-required publication still completes or fails, and MAP does not waive PTC's own required-publication obligations.

The six independently requestable product roles are `SCI-MAP:base_observation_map@1`, `SCI-MAP:response_bearing_observation_map@1`, `SCI-MAP:covariance_qualified_observation_map@1`, `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1`. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance.

| Role                             | Exact parent                                                                       | Required companions                                                                                                                   | Missing/incompatible companion                                                                                                                                                                                                                                                                       |
|----------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base observation                 | Exact realized PTC/AST inputs and complete MAP-local observation plan              | Complete MAP-local realized operator identity; honest upstream-response and covariance states                                         | Base signal may remain valid; state and cause are retained.                                                                                                                                                                                                                                          |
| Response-bearing observation     | Exact base-observation product named above                                         | Every named numerical response, with exact source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity       | This role fails or is unavailable before its required-product mutation; its base parent is unchanged.                                                                                                                                                                                                |
| Covariance-qualified observation | Exact base-observation product named above                                         | Every within-observation covariance block named by the role on exact rows                                                             | Missing blocks are unknown and block this role only; its base parent is unchanged.                                                                                                                                                                                                                   |
| Base coadd                       | Ordered set of exact complete base-observation bundles and one resolved coadd plan | Every base signal companion and honest response and covariance identities/statuses                                                    | Unavailable or basis-incompatible response and incomplete covariance are permitted with unchanged signal membership; coadd response becomes unavailable with exact cause.                                                                                                                            |
| Response-bearing coadd           | Exact base-coadd parent named above and the identical ordered signal-member set    | Every member response and their exact compatible source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity | If the base parent already exists, this stronger role is unavailable for that parent; it never selects a subset. A separately requested admission may reject an observation before mutation only while constructing its own separately identified base/signal parent. No hidden subset or zero fill. |
| Covariance-qualified coadd       | Exact base-coadd parent named above and the identical ordered signal-member set    | Every named within- and cross-observation block on exact stacked rows                                                                 | One missing required block blocks this role for that parent; no member subset is formed. A separately requested admission may reject before constructing its own identified base/signal parent. Unknown is never zero or independence.                                                               |

The target common grid may be requested and fixed before observation-map construction. The permitted placement is then exact centered-integer embedding. A smaller map may sit inside a larger common grid only when each dimension differs by a nonnegative even number of pixels; half the difference is the integer offset. The shifted reference pixel must identify the same world coordinate. An odd shape difference is incompatible and the complete observation is

rejected. There is no fractional shift, post-hoc pad, interpolation, reprojection, mosaicking, or implicit recentering in this ordinary branch. A later crop is a new row-selected product with its own identity, parent, and lineage; it cannot relabel its parent or make that parent retrospectively compatible.

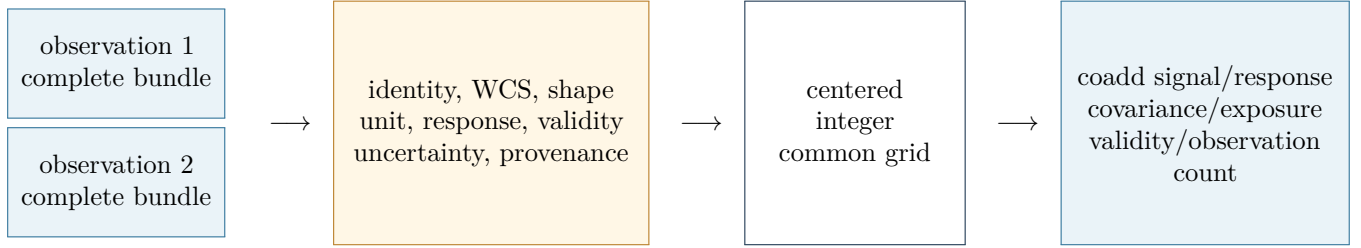


Figure 2: Complete-bundle admission and centered-integer placement. A failed fact required by the exact selected role rejects the observation before that role’s base/signal parent mutates; base-permitted response/covariance unavailability does not change base signal membership.

This restriction is the declared scope of the ordinary v0.2 coadd. It defines the approved preconstruction target-grid request and exact embedding, while leaving fractional shifting, resampling, interpolation, reprojection, and mosaicking to a separately authorized future contract.

The coadd-admission profile is `SCI-MAP:observation_coadd_admission@2`. It is a MAP-authored, VAL-governed aggregate profile with separate request, applicability, eligibility, and realization axes, distinct from the five-stage MAP lifecycle. The historical `SCI-MAP:observation_coadd_admission@1` remains immutable and is not an alias for this successor. It requires compatible quantity and nominal-beam convention, PTC route/product role, response class, complete AST WCS/grid, support policy, coefficient family, covariance disclosure, null/additive-reference state, required companions, lifecycle, and parentage. The coefficient family is `SCI-MAP:uniform_observation_coadd_coefficient@1`:  $u_{op} = 1$ , dimensionless, on every admitted observation row. It is an equal-observation coefficient and has no inverse-variance interpretation.

Writing  $\widehat{\mathbf{m}}_{\text{obs}}$  for the stacked signal on exact observation-row identities,  $B_{\text{out}}$  is the centered-integer averaging operator.  $T_{\text{obs}}$  stacks compatible fixed-state member response objects on the same domains, and  $\Sigma^{\text{obs}}$  contains every available within- and cross-observation covariance block. `SCI-MAP:base_coadd@1` permits unavailable or response-basis-incompatible response without changing numerical signal membership, but then coadd response is unavailable with exact cause; no smaller hidden membership or zero response is formed. `SCI-MAP:response_bearing_coadd@1` parents the exact base coadd with identical ordered members and requires every response object named by that role with compatible source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity. Missing a different unselected response family does not block it.

The equation  $C_{c,\text{out}} = B_{\text{out}}\Sigma^{\text{obs}}B_{\text{out}}^{\text{T}}$  is a complete numerical covariance statement only when every block named by `SCI-MAP:covariance_qualified_coadd@1` is available on that exact base membership. Otherwise the coadd publishes an explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable covariance state. Unknown blocks remain unknown, not zero or independence. If an existing base parent lacks a required companion, the stronger role is unavailable and never selects a subset. A separately requested stronger-role admission may reject an observation before building its own explicitly identified base/signal parent. The denominator is normalization, not precision. Correlated generalized least squares, which may need negative coefficients and a full  $\Sigma^{\text{obs}}$ , belongs outside this positive-coefficient branch.

## 7 Units, WCS, products, and downstream use

SCI-MAP receives its scale and unit convention only through the realized PTC product; there is no direct SCI-CAL fallback. It preserves the coordinates, exact frame metadata, geometry, and coordinate validity supplied by ALIGN/AST through `SCI-AST:rtc_output_grid_coordinates@1` for the same stable RTC sample and parent chain. It records the complete declared frame identity—including the actual reference system, equinox/epoch, time scale, topology, WCS, axes, signs, and handedness—and does not replace it with “J2000.” PTC owns transformed-sample identity and availability, producer-local validity, and MAP-facing coefficient/covariance facts. MAP owns the target grid and its own projection, coefficient-admission, boundary, support, and use conditions. VAL may represent and evaluate named rules; it does not author those policies.

The full binary64 typed or sidecar WCS is the lossless identity used for admission, common-grid comparison, and provenance. The WCS serialized in the FITS product is the representation ordinary scientific tools are expected to use for pixel-to-sky conversion. For an immutable finite MAP grid of  $n_x$  by  $n_y$  cells, the 0.1-arcsec representation-test population is the finite union of (a) every scientifically addressable integer pixel center  $(j+1, k+1)$  for  $0 \leq j < n_x$ ,  $0 \leq k < n_y$ , and (b) every outer-footprint cell vertex:  $(h + \frac{1}{2}, \frac{1}{2})$ ,  $(h + \frac{1}{2}, n_y + \frac{1}{2})$  for  $0 \leq h \leq n_x$ , together with  $(\frac{1}{2}, l + \frac{1}{2})$  and  $(n_x + \frac{1}{2}, l + \frac{1}{2})$  for  $0 \leq l \leq n_y$ . The second set includes all four outer corners and every required outer-boundary vertex. The complete population is preregistered from the full-precision typed WCS, grid shape, and row domain before the physical FITS WCS is generated. Both representations shall bind the same WCS-generation identity, celestial frame, axis order, signs, handedness, orientation, shape, and centered-integer reference-pixel relation. At every population member, each WCS is evaluated at the exactly corresponding center (zero-based memory index  $(j, k)$ , one-based FITS pixel  $(j+1, k+1)$ ). Separation is the great-circle angle  $\text{atan2}(\|\mathbf{u} \times \mathbf{v}\|, \mathbf{u} \cdot \mathbf{v})$  between unit direction vectors in that exact declared frame, evaluated with IEEE-754 binary64 round-to-nearest, ties-to-even operations and reported in arcseconds. The maximum over the complete finite population shall be at most 0.1 arcsec. If either WCS yields an invalid or non-finite direction at any preregistered member, the representation fails; the point is not dropped. A center is absent only when its scientific row was already absent before population generation; every listed outer-boundary vertex remains required. No smaller sampled subset or continuum/global-sky claim is authorized. The bound is representation authority only; this rationale gives it no physical interpretation.

The FITS WCS is sufficient for ordinary pixel-to-sky use at its declared serialization fidelity, while the sidecar retains the exact WCS identity used for admission, conformance, provenance, and coadd comparison. A tolerance never authorizes a different sky grid, fractional shift, interpolation, or reprojection.

| Science-user state                                | Interpretation and required action                                                                                                                                          |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Valid map; response available                     | Use only for the response class and processing lineage actually declared.                                                                                                   |
| Valid numerical map; response unavailable         | Preserve the honest limitation. The original bundle makes no unsupported response-dependent claim; later analysis may attach a new versioned response or corrected product. |
| Formal weight; empirical significance unavailable | Use only according to its declared role—normalization or formal precision—and do not call it significance.                                                                  |
| Covariance summarized or unavailable              | State meaning, domain, and limits. The map remains valid; later covariance work is a new bound product.                                                                     |
| Unsupported pixel                                 | Unavailable, not zero sky.                                                                                                                                                  |
| MAP-invalid pixel                                 | A downstream product may record separate use-local validity, but may not rewrite MAP-local base-product validity.                                                           |
| Coadd-incompatible observation                    | Entire observation bundle is excluded; no partial coadd.                                                                                                                    |
| WCS or unit incompatibility                       | Reject combination rather than silently convert, recenter, or relabel.                                                                                                      |

Downstream packages inherit the base/unfiltered MAP unit, response, covariance, validity, WCS, and provenance. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps  $Q$  normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims. FLT owns filtered response, covariance, and validity while preserving that parent; source and Beammap packages own their scientific inference. Pointing and

OOF may be registered consumers of the exact ordinary gridding operator when their inputs satisfy its contract, but their mode WCS/frame, source model, fitting, response interpretation, outputs, defaults, profiles, and qualification remain with their respective owners. No unsupplied mode profile becomes available.

## 8 Validation

Validation must keep algebra, software conformance, representation fidelity, observational performance, and production readiness separate. The required science program is:

1. analytic estimator tests for constants, unequal coefficients, and exact hand calculations;
2. exact one-hot half-open-boundary tests and explicit rejection of fractional projection until a successor family is authorized;
3. support, boundary, global-population, and unsupported-pixel tests;
4. complete-bundle compatibility and centered-integer coadd placement tests;
5. sequential/parallel conformance: floating results agree within a preregistered bound appropriate to operation count and conditioning, while identities, counts, masks, support, offsets, and product cardinality agree exactly;
6. synthetic point, extended, gradient, edge, and variable-coverage response tests; and
7. observational amplitude, shape, position, covariance, and repeatability checks against independent references.

No test, implementation inspection, reduction, or production qualification was executed for this authorship revision.

## 9 Scientific-owner decisions and remaining gates

The register in Appendix B preserves OD-001 through OD-009. All nine are decided for this successor candidate: eight use the approved 2026-08-28 dispositions and OD-008 carries the earlier one-hot decision. PTC-OD-010 supplies the versioned Registry architecture, and the separately approved uniform authority fills one exact family slot by reference. No open scientific choice remains within this incorporation scope.

This is still a candidate. Complete successor disposition by the scientific owner, fresh implementation-blind consistency review, exact final source and VAL binding, and freeze remain pending. Further revision should follow a normative discrepancy, owner disposition, new scientific evidence, or a demonstrated inconsistency rather than stylistic churn.

## References

- [1] *SCI-MAP v0.2/r0.2 Formal Scientific/Engineering Contract*, a view of this candidate's shared authority.
- [2] Frozen *SCI-MAP v0.1/r0.7.1*, SCI-PTC v0.1/r0.5, and SCI-AST v0.1/r0.3 exact authorities admitted by the approved author manifest.
- [3] The 2026-08-28 post-freeze MAP scientific-owner decisions and the exact 2026-09-07 Scope Brief approval.
- [4] The exact PTC-to-MAP and AST-footprint boundaries, the uniform coefficient family and handoff profile, the admitted uniform VAL Registry/source binding, and the exact MAP upstream-admission and observation-coadd records.
- [5] The content-bound 56-reference author manifest.

## A Compact rationale-to-contract crosswalk

The separate `CROSSWALK.md` carries the detailed routing. The formal contract remains normative; this appendix does not replace or paraphrase any requirement or prediction as authority.

| Formal authority | Science-team location           | Continuing formal home                                                             |
|------------------|---------------------------------|------------------------------------------------------------------------------------|
| REQ-001–010      | Executive summary; Sections 1–2 | Version, input, identity, projection, coefficient, lifecycle, admission.           |
| REQ-011–018      | Sections 2–3                    | Accumulators, valid-domain publication, response, absolute mode.                   |
| REQ-019–024      | Section 4                       | Conditional covariance, formal weight, nuisance state, statistical labels.         |
| REQ-025–035      | Section 5                       | Separate incidence, contribution, exposure, support, and MAP-local validity facts. |
| REQ-036–042      | Section 6                       | Complete bundles, centered-integer coadd, response/covariance, GLS boundary.       |
| REQ-043–048      | Section 7                       | Product identity, WCS, provenance, failure, publication.                           |
| REQ-049–052      | Sections 7–8                    | Base MAP parent, consumer boundary, fixed realization operator, claim layers.      |
| PRED-001–025     | Section 8                       | Full formal prediction inventory and exact evidence classes.                       |
| OD-001–009       | Section 9; Appendix B           | External scientific-owner decision ledger.                                         |
| SCI-MAP-CI-001   | Section 5                       | Resolved dimensionless classification and bounded normative amendment.             |

## B Compact scientific-owner decision register

All nine stable MAP decisions are decided. The complete evidence, resolution authority/date, blocked-claim boundary, and affected documents live in `SCIENTIFIC_OWNER_DECISION_LEDGER.md`.

| Decision/status                  | Approved disposition                                                                                                                                                                                                  | Binding consequence                                                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-MAP-OD-001<br><b>DECIDED</b> | Relative support is an operational policy, with no exposure, precision, significance, response, completeness, or physical-optimality meaning.                                                                         | Preserve the exact population, order statistic, thresholds, and policy identity.                                                                        |
| SCI-MAP-OD-002<br><b>DECIDED</b> | Independent experimentation is unrestricted; official identity and claims follow mathematical equivalence or versioned change control with appropriate evidence.                                                      | No permission process for research and no retroactive relabeling.                                                                                       |
| SCI-MAP-OD-003<br><b>DECIDED</b> | Every valid map discloses its known MAP-local realized operator. Upstream- conditioned and whole-chain responses remain separately typed and honest.                                                                  | Unsupported response-dependent claims remain unavailable; later estimates are new versioned products bound to the original.                             |
| SCI-MAP-OD-004<br><b>DECIDED</b> | Report actual covariance information, domain, assumptions, omissions, limitations, and lineage; a complete covariance object is not mandatory.                                                                        | Unknown is never zero or independence; later estimates have new identity.                                                                               |
| SCI-MAP-OD-005<br><b>DECIDED</b> | A complete logical observation bundle precedes coaddition; physical persistence follows the effective plan.                                                                                                           | In-memory and persisted/coadd-only routes use identical admission, arithmetic, identity, and provenance; required publication still completes or fails. |
| SCI-MAP-OD-006<br><b>DECIDED</b> | Pointing and OOF may reuse the exact ordinary MAP operator with conforming inputs.                                                                                                                                    | Mode WCS/frame, model/fitting, interpretation, outputs, defaults, profiles, and qualification remain with their owners.                                 |
| SCI-MAP-OD-007<br><b>DECIDED</b> | <code>coverage_cut</code> is finite and nonnegative: zero gives no relative cut; $(0, 1]$ is ordinary; $> 1$ requires an explicit expert request; bad or missing values fail before support construction or mutation. | Retain requested, effective, observation-resolved, applied, and realized states/values; no universal default.                                           |
| SCI-MAP-OD-008<br><b>DECIDED</b> | The one-hot lower-inclusive, upper-exclusive containing-pixel family remains the only ordinary projection.                                                                                                            | Boundary loss contributes nowhere; fractional projection remains deferred.                                                                              |
| SCI-MAP-OD-009<br><b>DECIDED</b> | A canonical target grid may be requested before construction; compatible maps use exact centered-integer embedding.                                                                                                   | A later crop is a new row-selected product; fractional resampling, reprojection, and mosaicking remain outside this contract.                           |

## C Document authority

Rationale revision r0.2 presents the targeted SCI-MAP v0.2 successor candidate. The shared source preserves 52 canonical requirement IDs and 25 canonical prediction IDs while incorporating the approved post-freeze and uniform-family decisions. The common modules are the normative authority; the formal and engineering documents are conformant rendered views. Scientific- owner disposition, independent consistency review, canonical activation of the supplied exact candidate source/VAL bindings, and freeze remain pending. No implementation, validation, availability, readiness, or production claim follows.